

4 EXPERIMENTS

We evaluate V4 on large-scale enzyme screening, bidirectional enzyme–reaction retrieval, and a literature-derived enzyme panel. The experiments address three questions: whether the model improves early enzyme recovery, whether its representations transfer across retrieval settings, and which training and inference choices account for the observed performance. Following FGW-CLIP (Zhou et al., 2026), we use EnzymeMap for screening and ReactZyme for bidirectional retrieval. Case1 assesses retrospective catalyst prioritization outside these benchmark candidate pools.

Implementation. We fit the same architecture and hyperparameter recipe separately to each benchmark training partition. The F3 retrieval layers are freshly initialized; the pretrained feature extractors remain frozen. Phase 1 uses anchor-balanced decoupled multi-positive learning with inverse temperature $\beta = 5$, batch size 512, and AdamW with learning rate 10^{-4} and weight decay 10^{-2} for 10 epochs. Phase 2 freezes F3 and trains the residual heads for 100 full-graph updates using uniform-positive cross-entropy, temperature 0.2, and identity-penalty weight 10. Its learning rate is 10^{-4} and weight decay is 10^{-3} . At inference, the training-reaction dictionary weight is $\alpha = 0.40$, protein fusion multiplier is $\nu = 3$, and residual multiplier is $c = 0.5$. We use neither ensembles nor hubness correction. The proposed method uses no added biological-label loss; this supervision is evaluated only as an ablation in Section 4.3. Primary results use seed 42. Downstream associations are matched across local comparisons, while pretrained representations and auxiliary annotations remain method-specific.

4.1 ENZYME SCREENING ON ENZYME MAP

Datasets. We adopt the processed EnzymeMap dataset and reaction-rule split distributed with CLIPZyme (Heid et al., 2023; Mikhael et al., 2024). The dataset contains 46,356 associations, 16,776 reactions, 12,749 enzymes, 2,841 EC identifiers, and 394 reaction rules. The released rule-based 80/10/10 split contains 34,427 training, 7,287 validation, and 4,642 test associations. The screening library combines EnzymeMap, BRENDA 2022_2, and UniProt 2022_01, and contains 261,907 candidate identifiers with sequences of at most 650 residues. We retain the released reactant/product roles and candidate library.

Following the released CLIPZyme notebook, the 4,642 test associations yield 1,551 distinct reaction strings; removing 30 that also occur in training leaves 1,521 screening queries. Positives are the corresponding protein identifiers in the raw EnzymeMap records intersected with the screening library, giving 4,544 positive query–candidate labels. We evaluate both the *full library* and a *training-ID-excluded library*, following the additional FGW-CLIP analysis. The latter contains 252,113 candidates and 1,337 queries with remaining positives. Identifier exclusion does not enforce sequence- or homology-disjointness.

Baselines and metrics. We compare with the reported CLIPZyme variants and FGW-CLIP, and evaluate the released CLIPZyme checkpoint locally using the same candidate axes, query eligibility, and positive labels as our model. This checkpoint was trained on the corresponding EnzymeMap partition. FGW-CLIP is a published reference rather than a local reproduction, as no verified executable release was available for these experiments. We report query-averaged BEDROC₈₅, BEDROC₂₀, EF5, and EF10; higher is better. BEDROC is shown as a percentage in Table 1. For n positives among N candidates, enrichment at fraction f is $EF_f = \sum_{j=1}^{\lfloor fN \rfloor} y_{(j)} / (fn)$, where $y_{(j)}$ is the ranked positive label. We preserve the released notebook’s zero-based BEDROC ranks and sorting convention. MRR is not a screening endpoint.

Results. Table 1 shows that V4 improves all four screening endpoints over the released CLIPZyme checkpoint and the reported FGW-CLIP values in both libraries. Full-library BEDROC₈₅ reaches 57.23%, compared with 44.69% for released CLIPZyme and 48.66% for FGW-CLIP, gains of 12.54 and 8.57 percentage points. EF5 increases from 14.09 and 14.91 to 16.86. After excluding training identifiers, BEDROC₈₅ remains 52.87% and EF5 is 16.42. The improvement therefore persists when training identifiers cannot be retrieved.

Table 1: EnzymeMap screening. BEDROC values are percentages; EF values are fold enrichments. † denotes published results from FGW-CLIP; CLIPZyme (released) is evaluated locally. The full and excluded libraries contain 261,907 and 252,113 candidates, respectively. Bold marks the best locally evaluated result in each column.

Method	Full library				Training-ID-excluded			
	B ₈₅	B ₂₀	EF5	EF10	B ₈₅	B ₂₀	EF5	EF10
CLIPZyme (ESM) [†]	36.91	53.04	11.93	6.84	–	–	–	–
CLIPZyme (CGR) [†]	38.91	57.58	13.16	7.73	–	–	–	–
CLIPZyme [†]	44.69	62.98	14.09	8.06	39.13	58.86	13.40	7.81
FGW-CLIP [†]	48.66	66.69	14.91	8.18	45.14	61.43	13.57	7.61
CLIPZyme (released)	44.69	62.98	14.09	8.06	39.13	58.86	13.40	7.81
V4	57.23	75.54	16.86	8.96	52.87	72.64	16.42	8.79

Table 2: ReactZyme partitions used by every local method. Train/validation/test columns count association rows; $|R_{\text{test}}|$ and $|E_{\text{test}}|$ are the complete test catalogs, whose query/candidate roles swap by direction.

Split	Train	Val.	Test	$ R_{\text{test}} $	$ E_{\text{test}} $
Reaction-Sim	147,393	16,377	14,689	386	14,688
Enzyme-Sim	152,748	16,972	8,739	1,573	8,734
Time	149,554	16,618	12,287	2,634	12,277

4.2 BIDIRECTIONAL RETRIEVAL ON REACTZYME

Datasets. ReactZyme associates SwissProt enzymes with Rhea reactions and provides reaction-similarity, enzyme-similarity, and temporal splits (Hua et al., 2024). The released resource comprises 178,463 associations, 178,327 enzymes, and 7,726 reaction identifiers. We preserve the existing training, validation, and test partitions, with retained counts in Table 2. Each split contains 178,459 retained association rows. The temporal split uses the released 31 December 2010 cutoff. All local methods use the same complete test query and candidate catalogs in each direction. ReactZyme supplies flattened reaction participants in this materialization; its self-reaction encoding does not recover physical reactant/product roles.

Baselines. We report two comparison groups. Published references include FGW-CLIP’s EC Mode/Max variants and TIGER’s main ESM2Text/ProtT3 variants (Zhou et al., 2026; Zhang et al., 2026). Local comparisons retrain the released MLP, contrastive MLP, Transformer, and Bi-RNN families on the same downstream partitions as V4. For a compact controlled presentation, we show their ESM2/MAT-2D configuration throughout, together with a corrected contrastive-label sensitivity, an EnzGFM backbone control, and original Horizyn (Rocks et al., 2026). This feature pair is fixed across families rather than selected by test performance. The public-model campaign covers 126 completed configurations; its complete run inventory is retained separately.

The classifier baselines use four deterministic negative corruptions per positive, excluding known positives, because the original negative files are unavailable. They select checkpoints by validation BCE, with at most 200 epochs and patience 30. The corrected contrastive row resolves opposite label conventions in the released BCE and contrastive branches. Horizyn uses its original MLP towers, fingerprints, ProtT5 features, and MLNCE loss, with a participant-set reaction adapter; it trains for at most 100 epochs and selects by validation MRR. The EnzGFM row is a frozen-backbone control with adapted preprocessing. These comparisons match downstream associations but differ in inputs, objectives, and optimization budgets.

Metrics. For query q , let $P(q)$ denote its positive candidates and $\text{rank}_q(p)$ their one-based ranks. We use the benchmark implementation’s query-averaged all-positive reciprocal rank,

$$\text{MRR}_{\text{all}} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{|P(q)|} \sum_{p \in P(q)} \frac{1}{\text{rank}_q(p)}. \quad (1)$$

Table 3: ReactZyme retrieval. The upper group contains published MRR ($\dagger$); the lower group uses the shared all-positive evaluator in Eq. 1. Released-family rows use ESM2/MAT-2D; the EnzGFM control uses MAT-2D. Training budgets and pretrained inputs differ. Bold marks the best local value; display precision does not resolve very small differences.

Method	Reaction-Sim		Enzyme-Sim		Time	
	R→E	E→R	R→E	E→R	R→E	E→R
FGW-CLIP (EC Mode) $\dagger$	0.3181	0.3722	0.5253	0.8683	0.3394	0.5222
FGW-CLIP (EC Max) $\dagger$	0.3113	0.3804	0.5300	0.8581	0.3392	0.5229
TIGER (ESM2Text) $\dagger$	0.3190	0.5180	0.5920	0.9560	0.3660	0.6900
TIGER (ProtT3) $\dagger$	0.3370	0.4720	0.5790	0.9400	0.3720	0.6830
MLP	0.1057	0.2187	0.2155	0.7080	0.1046	0.2849
Contrastive MLP	0.0406	0.2398	0.1198	0.6667	0.0405	0.2550
Transformer	0.1220	0.2314	0.2163	0.7260	0.1419	0.3943
Bi-RNN	0.1510	0.2835	0.2736	0.7970	0.1274	0.3752
Corrected contrastive	0.0638	0.2431	0.1047	0.6316	0.0410	0.2412
EnzGFM control	0.1354	0.2965	0.2864	0.7989	0.1525	0.3867
Horizyn (adapted)	0.4318	0.5348	0.6997	0.9770	0.5982	0.8406
V4	0.4004	0.5344	0.6662	0.9714	0.5383	0.7807

Local scores use FP32 ranking and stable candidate-index tie breaking. First-positive MRR is a different quantity. Because published descriptions such as FGW-CLIP’s formula use the first positive, and predictions are unavailable to reconcile the implementation, published values are kept separate from the common-evaluator results.

Results. V4 achieves R→E/E→R MRR of 0.4004/0.5344 on Reaction-Sim, 0.6662/0.9714 on Enzyme-Sim, and 0.5383/0.7807 on Time (Table 3). These exceed the displayed main-method literature values, subject to the metric distinction above. However, original Horizyn exceeds V4 in all six locally evaluated cells. Its R→E MRR is 0.4318 on Reaction-Sim and 0.5982 on Time, compared with 0.4004 and 0.5383 for V4. Thus V4’s strong screening performance does not imply dominance in bidirectional retrieval. The larger Horizyn training budget also prevents isolating architecture as the cause of this ordering.

4.3 ABLATION STUDIES

Training stages. We compare the same target-specific backbone checkpoints before refinement, after the complete phase-2 pipeline, and after adding biological supervision (Table 4). The inference fusion adjustment is fixed across all columns. Phase 1 uses backbone cosine similarity; phase 2 adds both residual refinement and dictionary scoring, improving all 14 benchmark endpoints. This comparison measures their joint contribution and does not isolate refinement from the dictionary. Adding biological labels yields mixed, mostly small effects, so the proposed method omits this auxiliary loss.

Inference settings. We vary the dictionary weight and residual multiplier with trained checkpoints fixed (Figure 1). Increasing α from 0.25 to 0.40 at $c = 1$ improves Reaction-Sim E→R MRR from 0.5238 to 0.5377. Increasing it further to 0.50 reduces R→E MRR on all three splits, while improving EnzymeMap EF5/EF10 in both libraries and reducing BEDROC₈₅. At $\alpha = 0.40$, reducing c from 1 to 0.5 also lowers screening BEDROC₈₅. These results identify a trade-off between very early recognition, broader enrichment, and retrieval direction; the final settings are not uniformly optimal for every endpoint.

Contrastive temperature. An 18-epoch EnzymeMap study compares inverse temperatures $\beta = 5$ and $\beta = 10$ with paired seeds 17, 42, and 73 and the remaining recipe fixed (Figure 2). Lower inverse temperature improves BEDROC₈₅, BEDROC₂₀, and EF5 for every seed in both libraries; seed 73 accounts for the two negative EF10 differences. Mean full-library BEDROC₈₅ increases from 0.4940 ± 0.0350 to 0.5217 ± 0.0375 , where dispersion is sample standard deviation across

Table 4: Matched training-stage comparison. All values are raw metric units; BEDROC is not multiplied by 100 here. Phase 2 is the proposed method; phase 2 + Bio is an ablation. The same backbones, fusion adjustment, and candidate pools are used throughout. Results use seed 42.

Dataset / setting	Metric	Phase 1	Phase 2	Phase 2 + Bio
ReactZyme / Reaction-Sim	E→R MRR	0.493650	0.534366	0.534777
ReactZyme / Reaction-Sim	R→E MRR	0.356631	0.400381	0.400412
ReactZyme / Enzyme-Sim	E→R MRR	0.944501	0.971355	0.967373
ReactZyme / Enzyme-Sim	R→E MRR	0.625444	0.666249	0.666234
ReactZyme / Time	E→R MRR	0.696620	0.780669	0.800614
ReactZyme / Time	R→E MRR	0.427677	0.538285	0.538317
EnzymeMap / Full library	BEDROC ₈₅	0.545263	0.572337	0.572096
EnzymeMap / Full library	BEDROC ₂₀	0.736656	0.755433	0.755322
EnzymeMap / Full library	EF5	16.739662	16.859042	16.860921
EnzymeMap / Full library	EF10	8.867905	8.958692	8.956501
EnzymeMap / Training-ID-excluded	BEDROC ₈₅	0.518774	0.528734	0.528464
EnzymeMap / Training-ID-excluded	BEDROC ₂₀	0.716511	0.726403	0.726275
EnzymeMap / Training-ID-excluded	EF5	16.362318	16.418633	16.435718
EnzymeMap / Training-ID-excluded	EF10	8.700613	8.793103	8.803076

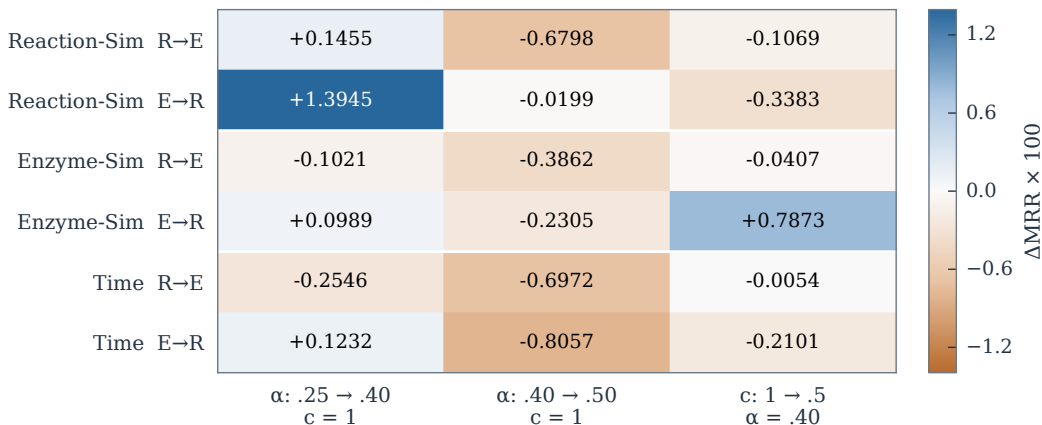


Figure 1: Inference sensitivity on ReactZyme. Entries are changes in MRR multiplied by 100 at fixed checkpoints. Each column changes one coefficient; the common color scale is linear, with blue/orange denoting positive/negative changes. Rows retain both directions for every split.

three seeds. This supports a temperature effect within the tested objective, but does not isolate anchor balancing against MLNCE or replicate the final 10-epoch V4 recipe. The 24 metric comparisons are correlated.

Biological-supervision ablation. We augment the proposed phase-2 objective with relative EC, cofactor, and bond-change losses, with coefficients (1, 1, 1), margin 0.1, and a fixed divisor of three. This ablation, denoted V4+Bio, adds no model parameters. We hold F3 and inference fixed and compare correct phase-2 labels with shuffled labels and removal of EC, cofactor, or transformation supervision. Removing a family preserves the other coefficients and divisor of three, therefore also reducing total regularization. The effects are mixed (Figure 3). Full-library BEDROC₈₅ is 0.572337 without added supervision, 0.572096 with correct labels, and 0.572093 with shuffled labels. All five controls slightly reduce both BEDROC endpoints in both libraries. On ReactZyme, the larger Enzyme-Sim/Time E→R changes are sensitive to near-equal scores: optimistic and pessimistic rankings within a 10^{-6} score band overlap substantially between V4 and V4+Bio. The Time increase from 0.7807 to 0.8006 therefore does not alone establish robustly improved discrimination. These bounds measure numerical sensitivity, not sampling uncertainty.

A separate study applies biology during F3 training with paired seeds 42, 43, and 44 and leaves phase 2 unannotated. Weight 1 increases mean full-library BEDROC₈₅ by 0.0103 but lowers mean

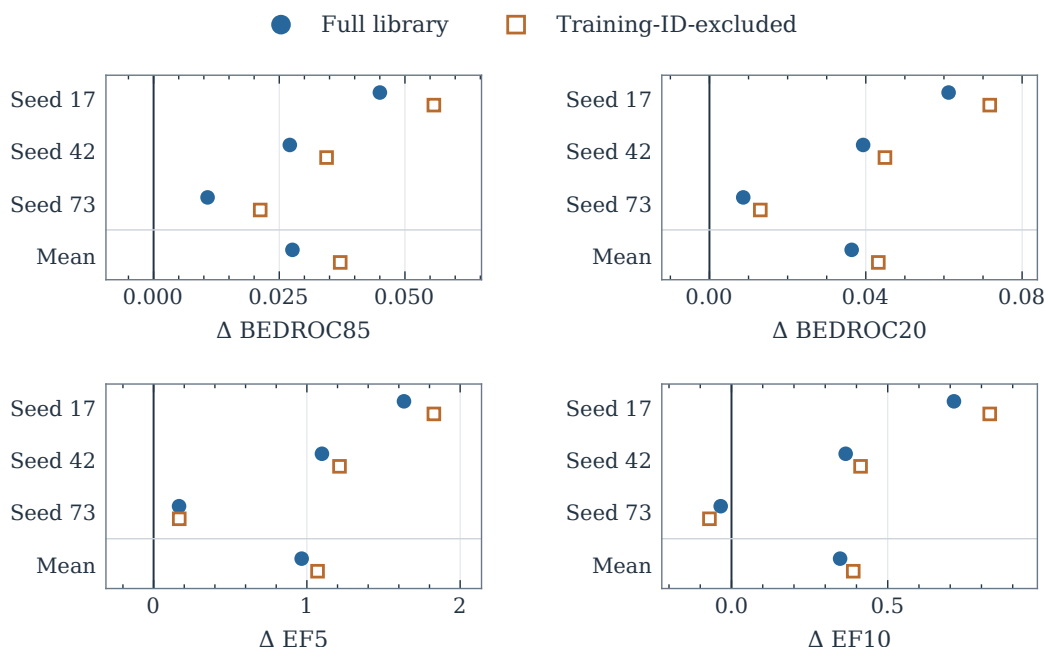


Figure 2: Paired temperature effects on EnzymeMap: $\beta = 5$ minus $\beta = 10$ at 18 epochs. Circles show the full library and open squares the training-ID-excluded library. Every seed and their arithmetic mean are shown. BEDROC differences are unscaled; each metric has its own horizontal axis. Mean markers do not represent confidence intervals.

Reaction-Sim R→E	+0.0031	+0.0147	-0.1254	+0.0030	+0.0006
Reaction-Sim E→R	+0.0412	+0.1237	+0.0710	+0.0404	-0.0264
Enzyme-Sim R→E	-0.0015	-0.0264	-0.0008	-0.0015	+0.0045
Enzyme-Sim E→R	-0.3983	-0.0465	-0.1550	+0.0448	-0.3926
Time R→E	+0.0032	+0.0591	+0.0039	+0.0033	-0.0005
Time E→R	+1.9945	+0.9596	+0.9125	+3.5192	+2.3369
	All labels	Shuffled labels	Without EC	Without cofactor	Without transform.

Color scale: $\Delta \text{MRR} \times 100$ (ranging from -1 to 1)

Figure 3: Phase-2 biological supervision on ReactZyme. Values are $100\Delta\text{MRR}$ relative to V4. Colors share a symmetric logarithmic scale, linear within ± 0.001 displayed units; printed values are untransformed. Small effects remain visible without independent row scaling. The largest E→R changes require the near-tie qualification in the text.

EF5 by 0.1514; only two of three seeds improve BEDROC₈₅. EF5 decreases in the excluded library for all three seeds. Weight 0.1 improves excluded-library EF10 for all seeds, but not every other endpoint. Both training stages therefore show metric-dependent biological-loss effects. The available sensitivities do not replace matched removals of the dictionary, phase 2, or learned residue views.

Table 5: Case1 primary-paper recovery at rank 25. Unique recovery is out of 12 positives among 123 sequences; entry recovery is out of 15 among 144 resolved entries. AUROC uses the broader 81/42 conditional workbook labels. The EnzymeMap model is a transfer diagnostic.

Model	Unique @25	Entries @25	AUROC
Native F3	9/12	10/15	0.6008
V4	10/12	10/15	0.6129
V4 (EnzymeMap)	6/12	6/15	0.4706

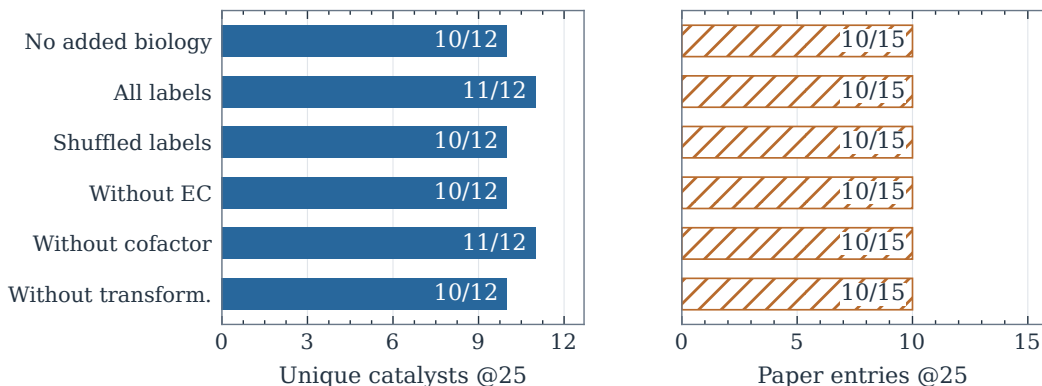


Figure 4: Case1 phase-2 controls using the same Reaction-Sim model lineage. Left: unique-sequence recovery at 25; right: original-entry recovery at 25. Blue filled and orange hatched bars distinguish the evaluation units, with their positive denominators printed explicitly. The one-sequence gain does not increase entry-level recovery.

4.4 CASE1: RECOVERY OF LITERATURE-SUPPORTED TAGATOSE CATALYSTS

Reaction and candidate panel. Case1 concerns direct C4 epimerization of free D-fructose to D-tagatose. We assemble a targeted panel from the literature, including engineered UxaE variants and characterized tagatose 4-epimerases (Shin et al., 2020; Xia et al., 2024; Chen et al., 2024; 2025; Zhao et al., 2025; Wang et al., 2025; Zhu et al., 2026; Guo et al., 2026). The catalogue contains 145 entries; one construct, H017 (KoT4E M6), lacks a fully recoverable sequence. We therefore rank 144 resolved entries, representing 123 unique sequences. The primary-paper evidence set contains 12 unique sequences or 15 entries. Broader workbook annotations identify 81 active sequences and 42 with conditional non-detection evidence; these heterogeneous labels are used only for a secondary conditional AUROC. The curated candidate ranking is not treated as an activity label.

Evaluation. We use the Reaction-Sim-trained model as the primary deployment without fitting its retrieval parameters to Case1 labels. Recovery is measured against primary-paper positives in both candidate views. Uniform random ranking would recover an expected $25(12/123) = 2.44$ unique positives or $25(15/144) = 2.60$ positive entries at rank 25. These expectations are descriptive references, not confidence intervals. The case was examined during development and is a retrospective literature evaluation, rather than an untouched prospective wet-lab experiment.

Results. V4 recovers 10/12 unique paper-supported catalysts at rank 25, compared with 9/12 for its native F3 encoder before phase 2 (Table 5), and recovers 10/15 positives in the 144-entry view. The biological-supervision ablation recovers 11/12 unique catalysts while retaining 10/15 entry-level recovery. The additional unique hit is TsT4Ease WT, which moves from rank 26 to 25. Cofactor removal retains this gain, whereas label shuffling and transformation removal do not (Figure 4). Conditional AUROC changes only from 0.6129 to 0.6138. V4 retrieves no primary-paper catalyst among its first five unique sequences and three among its first ten, showing that top-25 recovery does not imply strong very-early prioritization.

The EnzymeMap-trained model recovers 6/12 unique positives and has conditional AUROC 0.4706, indicating weaker transfer despite its strong screening benchmark performance. Furthermore, four of the ten positives recovered by Reaction-Sim V4 have identified training homologs with at least 90% identity, and both query molecules co-occur in larger training participant sets. Frozen-backbone exposure is not ruled out. Case1 thus supports prioritization within this targeted panel, with limited evidence for distant catalytic generalization.

Scope of the evidence. The experiments establish improved EnzymeMap early retrieval under two released screening protocols, competitive published-reference ReactZyme scores, and retrospective recovery of documented catalysts. They also identify a stronger matched-data ReactZyme comparator and mixed effects from added biology. The final recipe followed repeated exploratory test inspection; primary results are single-seed estimates. Replication of the final recipe and evaluation on a prospectively specified external reaction panel are needed to establish broader generalization.

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